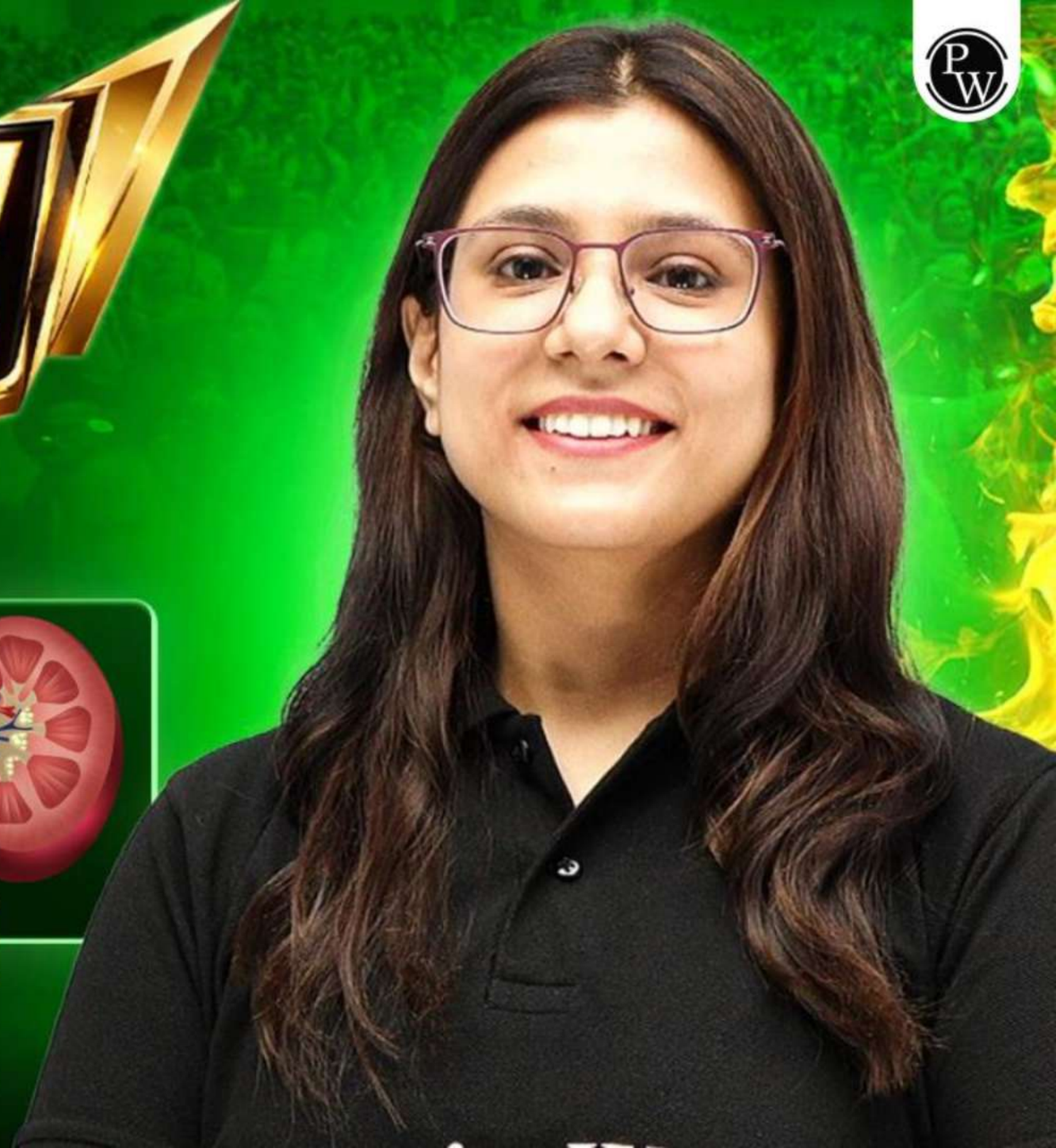
VICTORY 2027

# The Excretory System



LECTURE-2 BIOLOGY

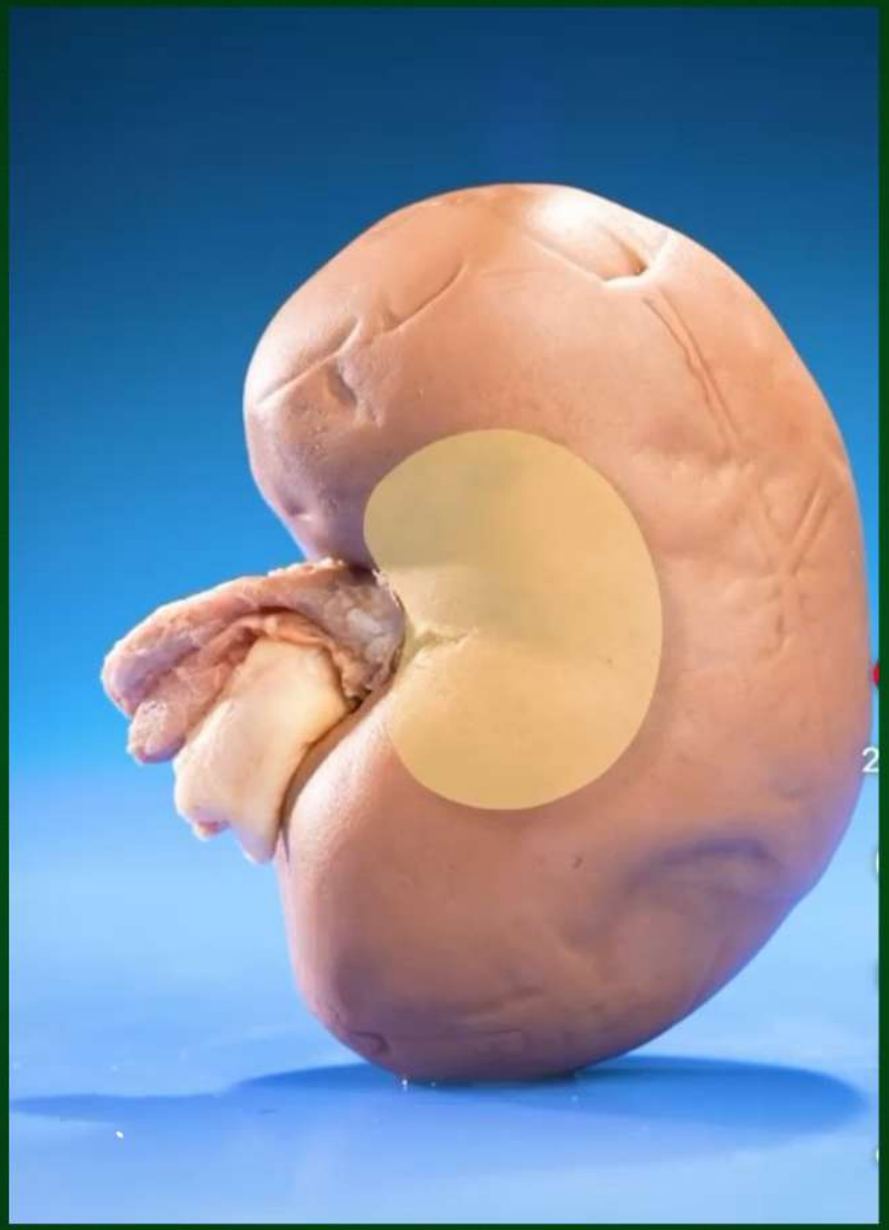
Upasana Ma'am

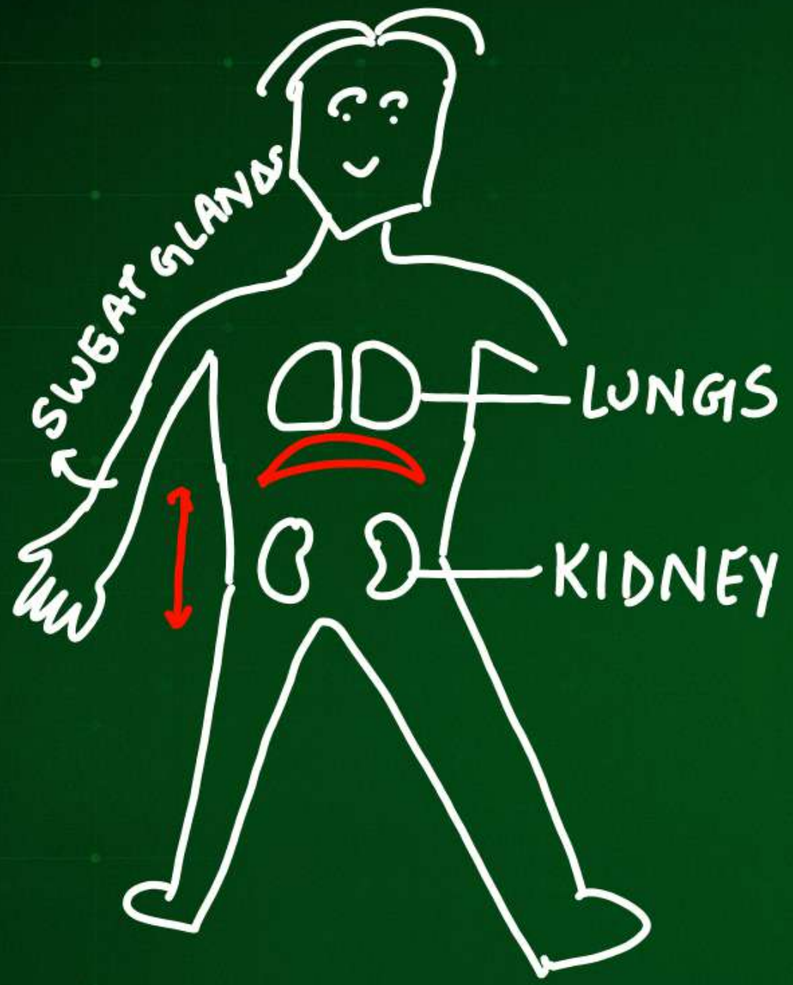


# Topics To Be Covered

1. Kidney
2. The Excretory Organs
3. Internal Structure of the Kidney


HELLO
NINJAS
忍者参上
影の中で戦う
忍者
忍術最強
忍者
忍者参上



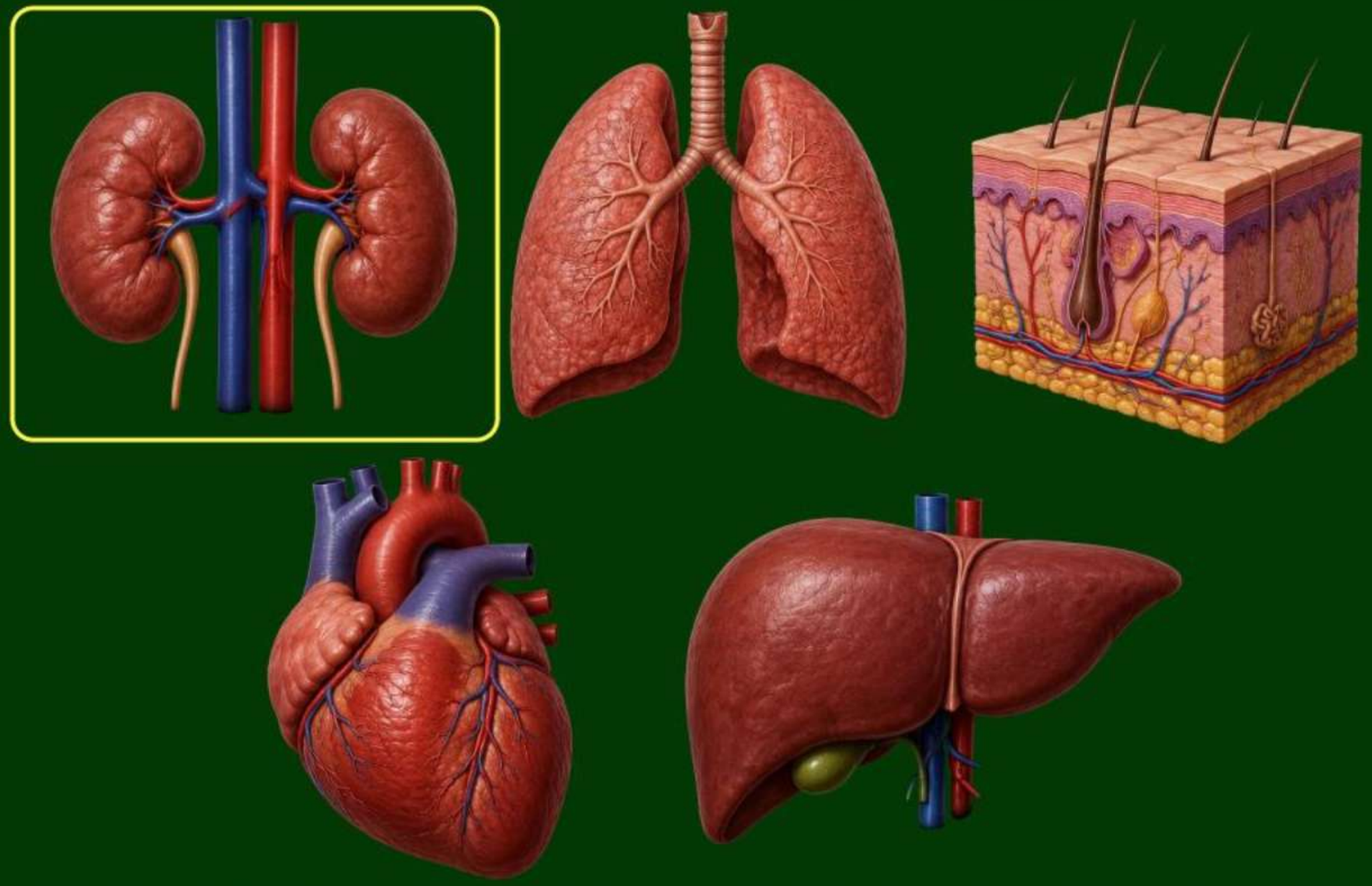



## Waste Products

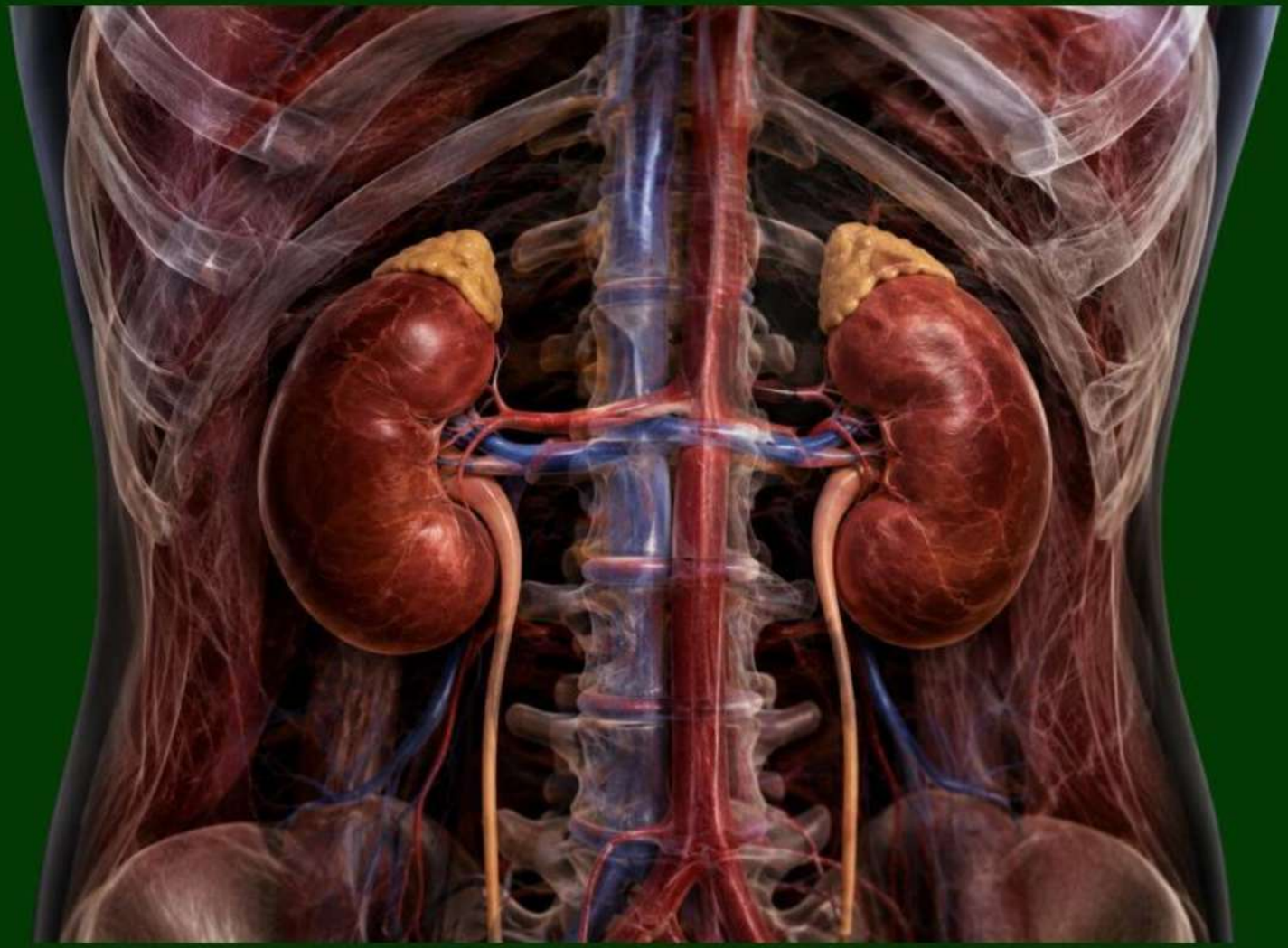
- → $CO_2$
- → Extra water, salts, ions
- → UREA (main waste)
- → Ammonia, Uric acid (small Quantities)
- → Bile Pigment

Nitrogenous waste

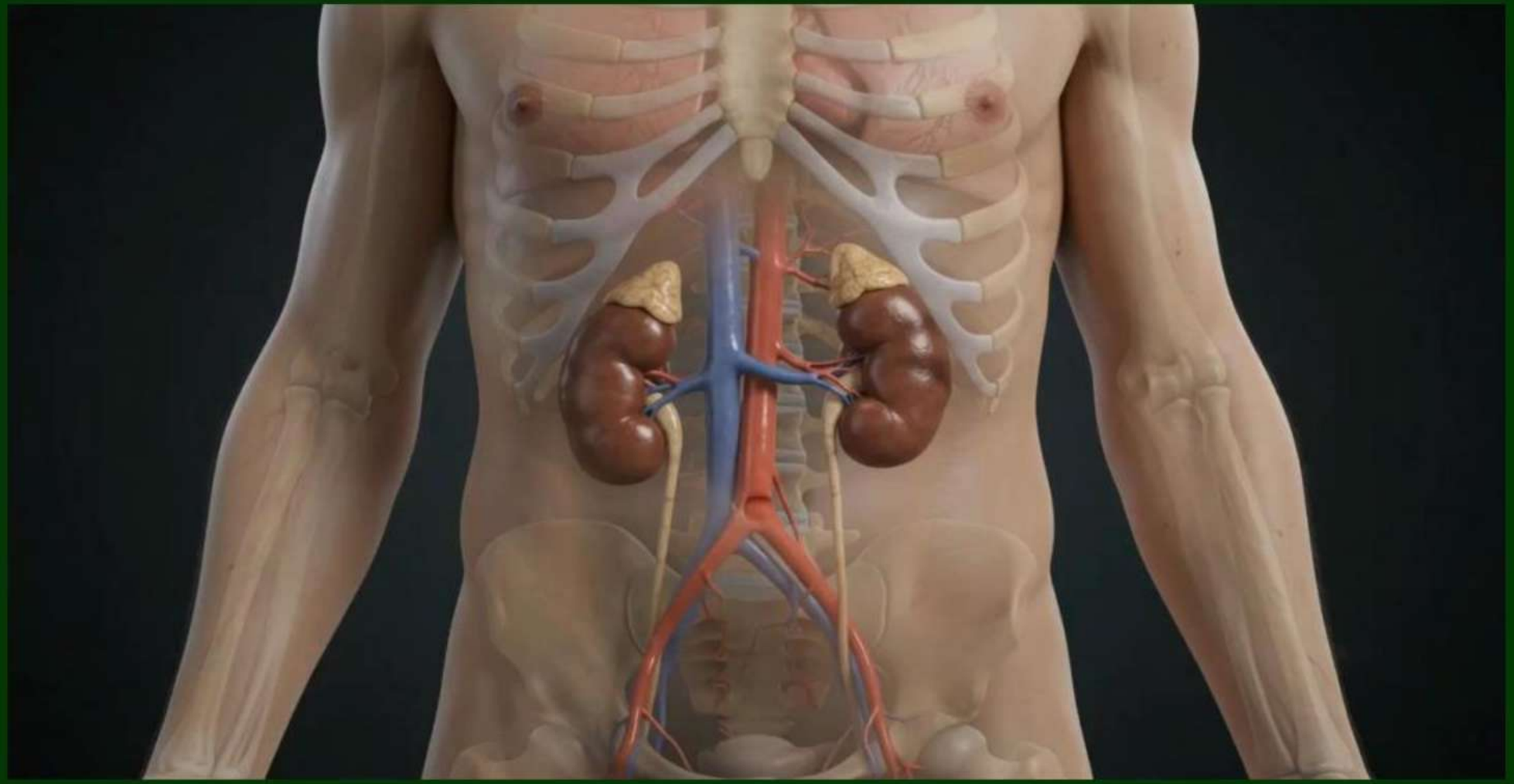
# WHICH OF THESE IS THE MAIN ORGAN OF THE URINARY/EXCRETORY SYSTEM?



# KIDNEY



# KIDNEY



# KIDNEY



# THE EXCRETORY ORGANS

## 1. KIDNEYS

- Primary excretory organs eliminating nitrogenous waste from blood through urine

Urea, $NH_3$, Uric acid

# THE EXCRETORY ORGANS

## 2. SKIN

- Sweat excreted by sweat gland of skin excretes out water and NaCl along with small amount of urea and lactic acid.
- Sweat gland pass out sweat only for cooling are not for excretory.



# THE EXCRETORY ORGANS

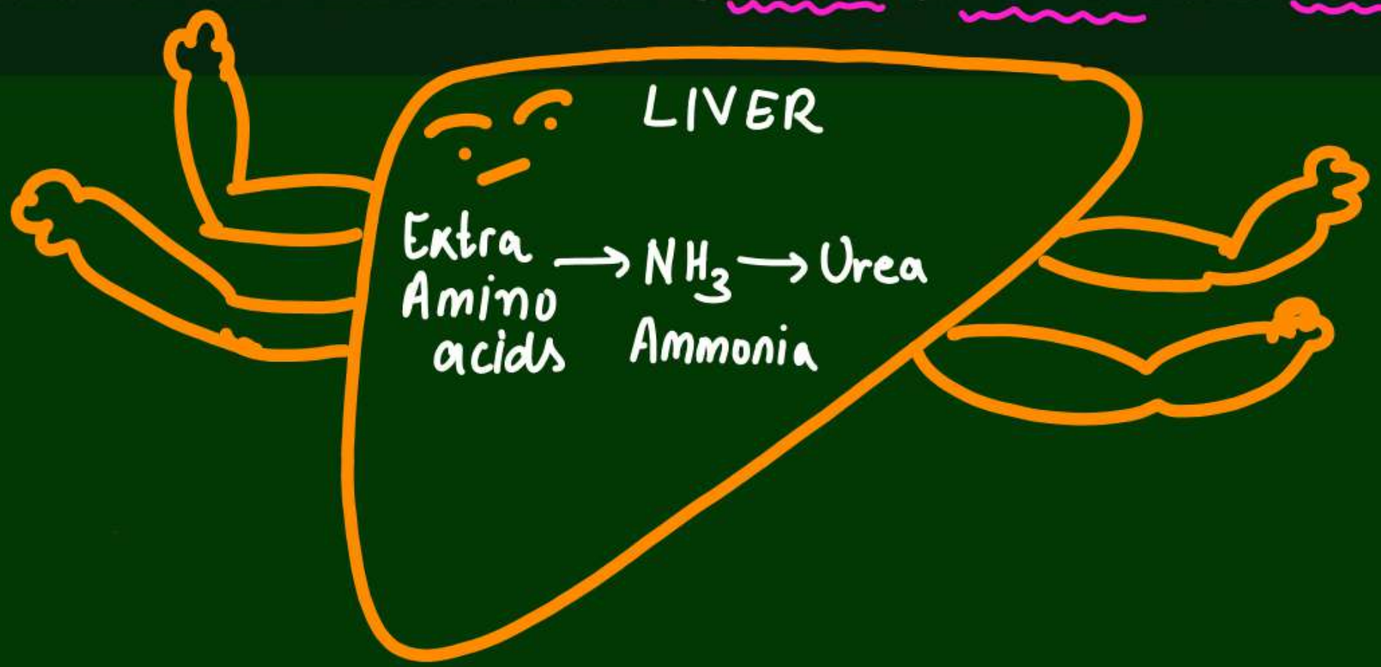
## 3. LUNGS

- Excrete out $CO_2$ from body through expired air.

# THE EXCRETORY ORGANS

## 4. LIVER

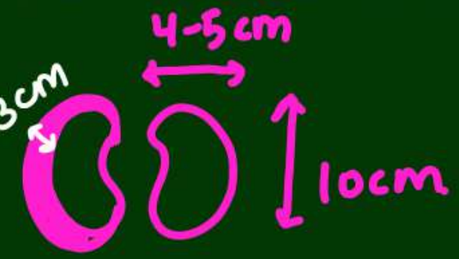
- Detoxification of ammonia by converting it into urea.
- Also break down bad cholesterol, alcohol, nicotine and drugs.



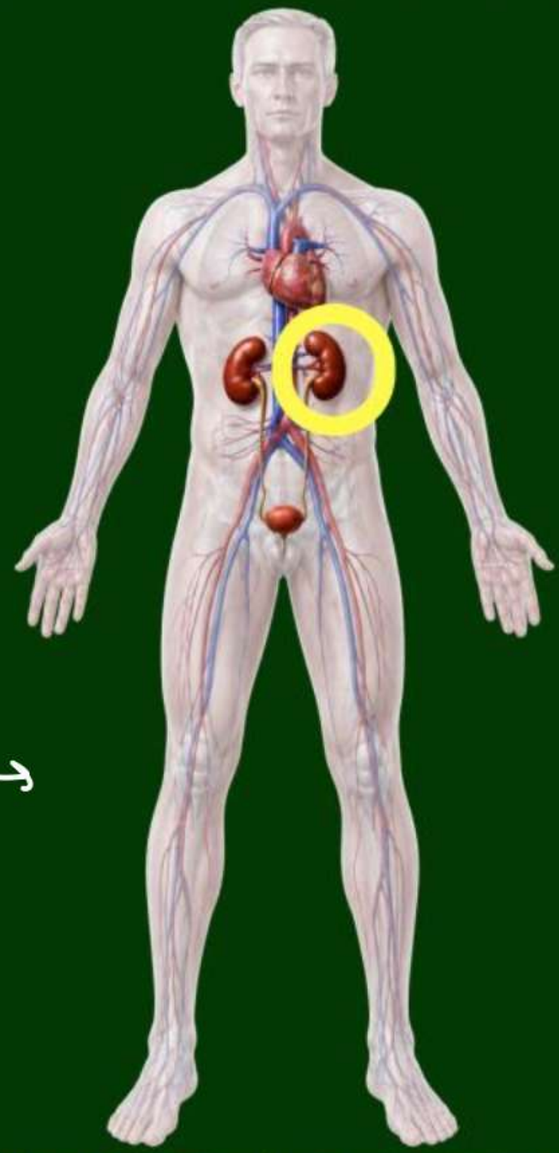

Protein
↓
Amino acids

# KIDNEY

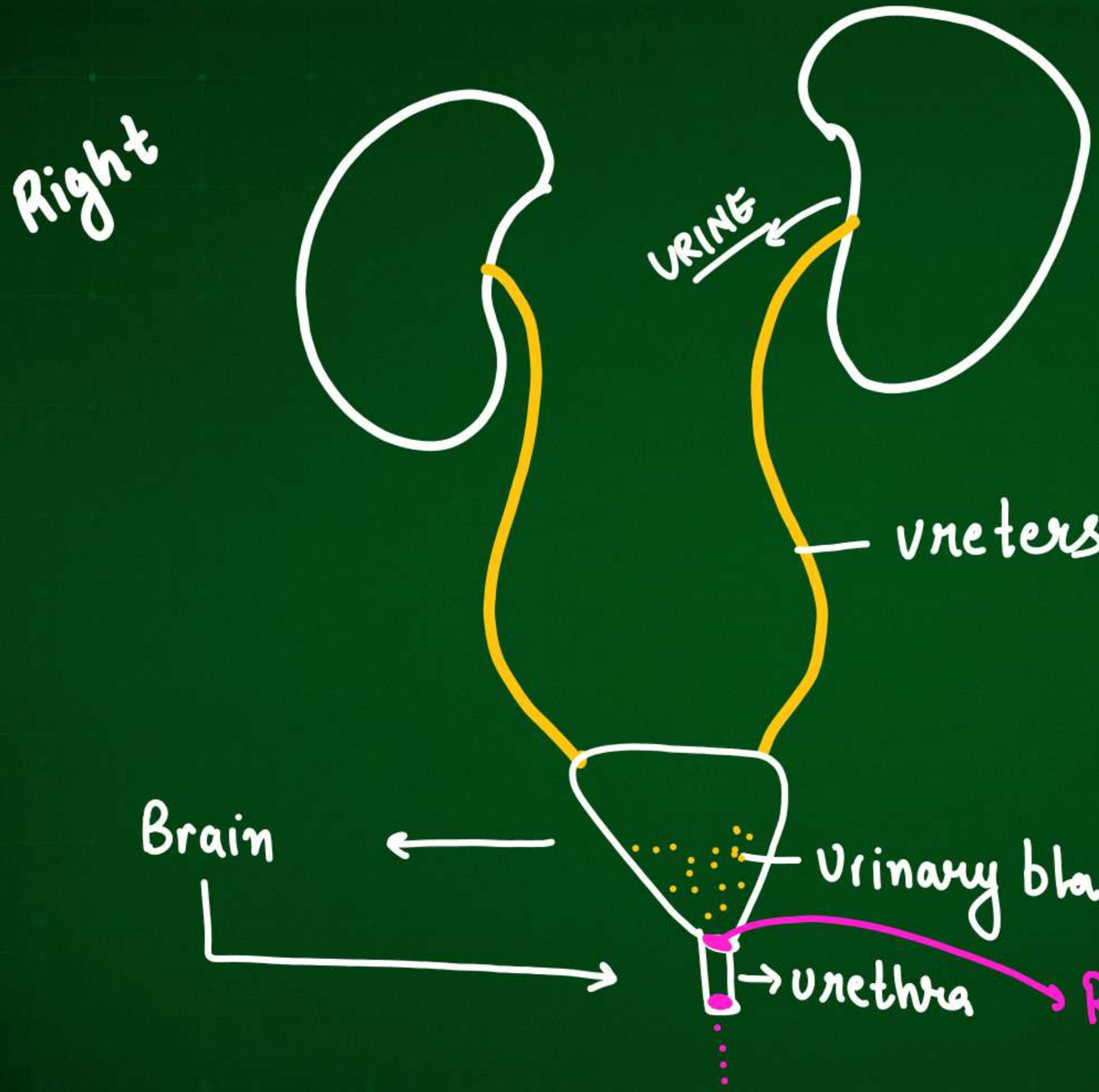





- 2 bean shaped organ ✓
- 10cm long and 6cm wide
- Located on either side of backbone and protected by last 2 ribs. (floating ribs) dorsally
- Ureter arises from the notch (hilum) in the median surface of each kidney and connects behind with the urinary bladder in lower abdomen part.



10 ↓ 5 ↓ 3




## The Excretory System

- Paired Kidney
- Paired Ureters
- One Urinary Bladder
- One Urethra.

# Colour of URINE




Haem
Globin (protein)

Haem → Biliverdin (green) → Bilirubin (yellow)

Bile Pigments.

Bile Juice from Gall Bladder → Small Intestine

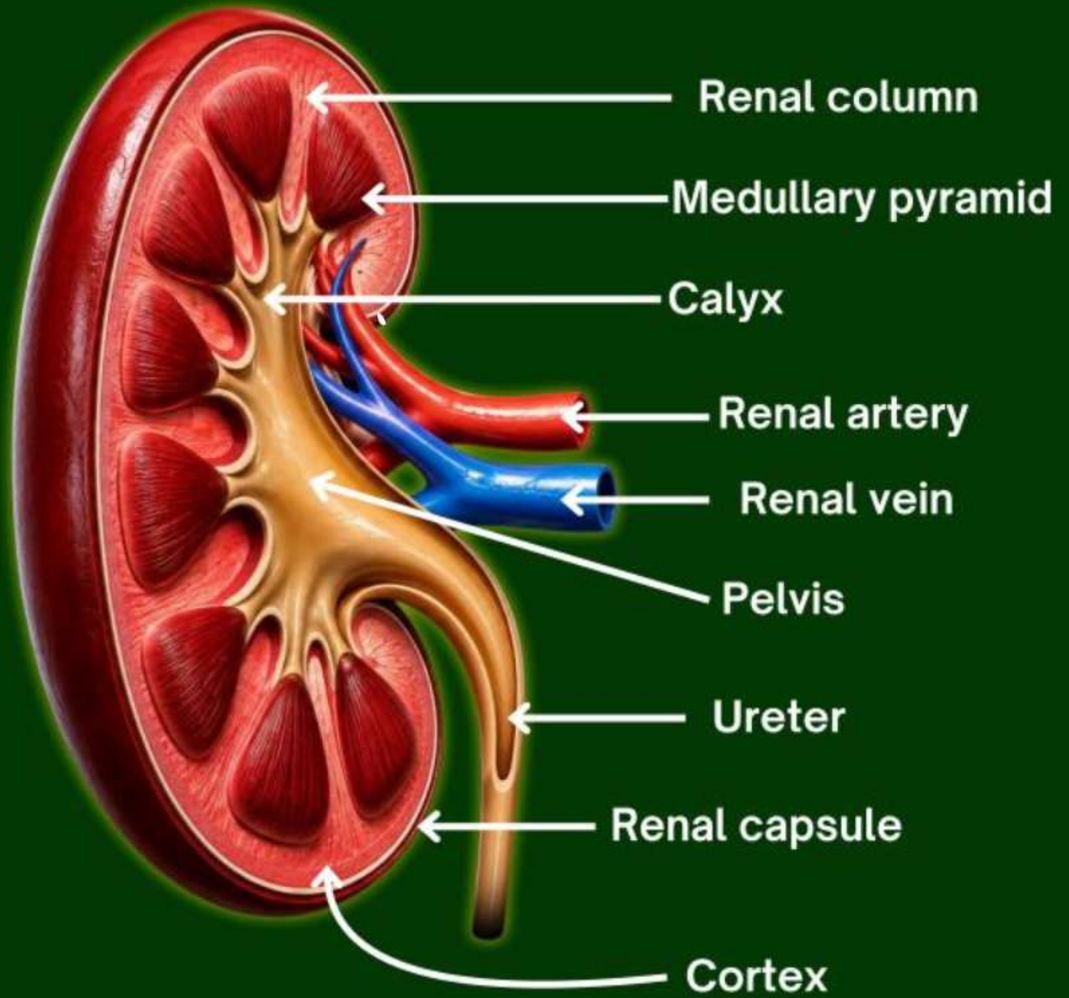
Gall Bladder

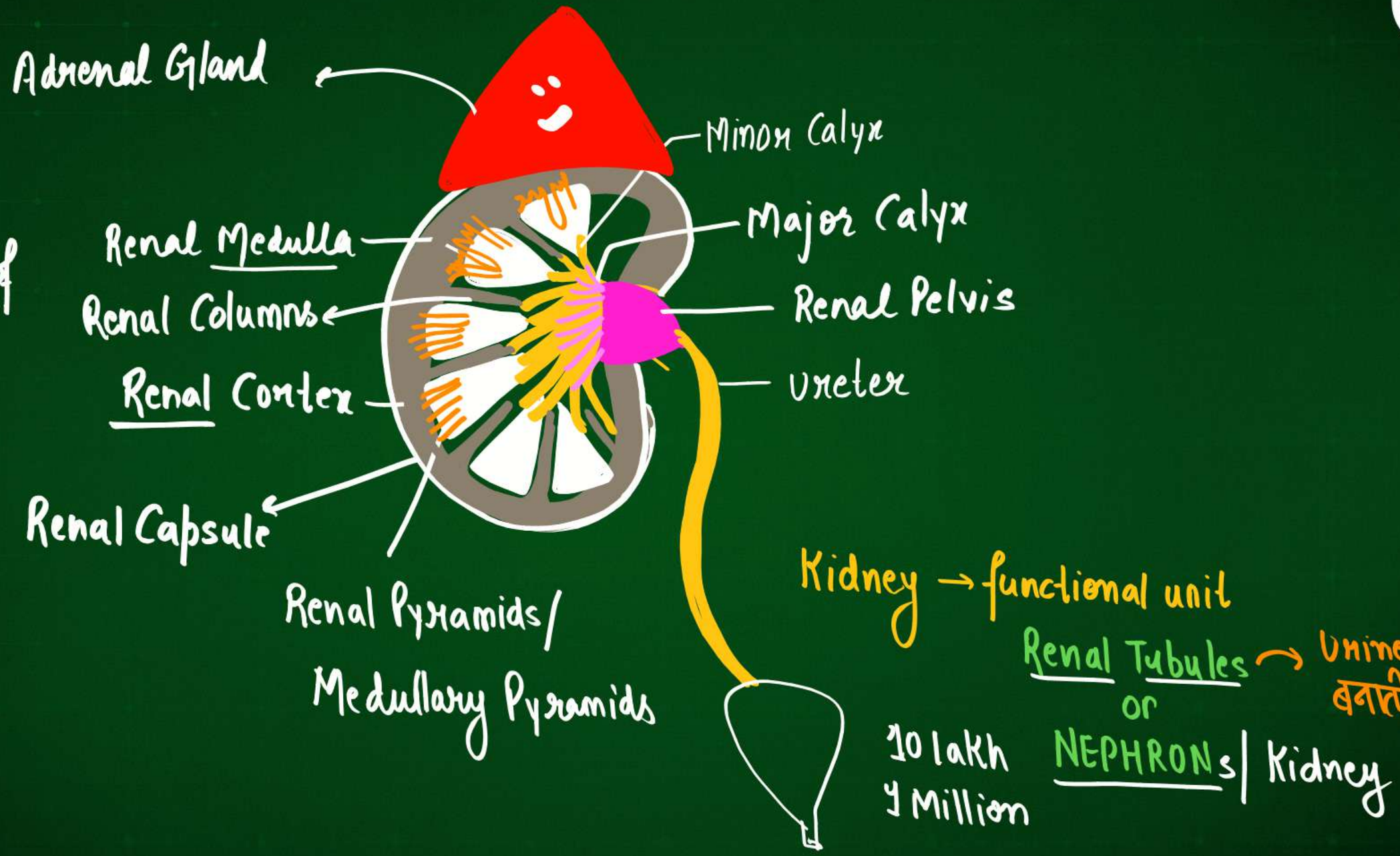
faecal matter [Brown]

Urine [Urochrome] ★

Pigment (Yellow colour)

# KIDNEY


Renal column
Medullary pyramid
Calyx
Renal artery
Renal vein
Pelvis
Ureter
Renal capsule
Cortex




Kidney → functional unit

Renal Tubules → Urine बनाती है।
or
NEPHRONs/ Kidney

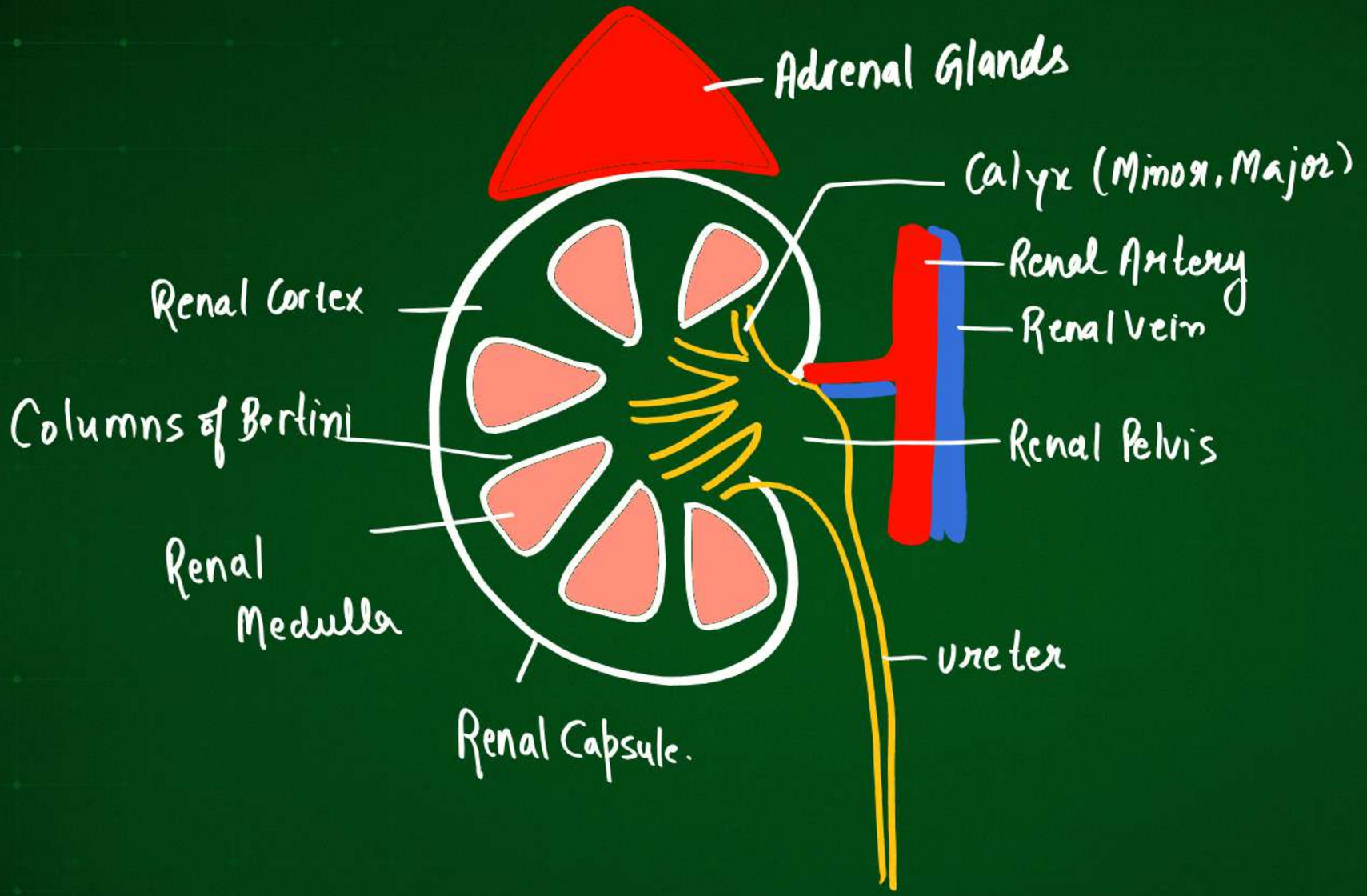
10 lakh
1 Million

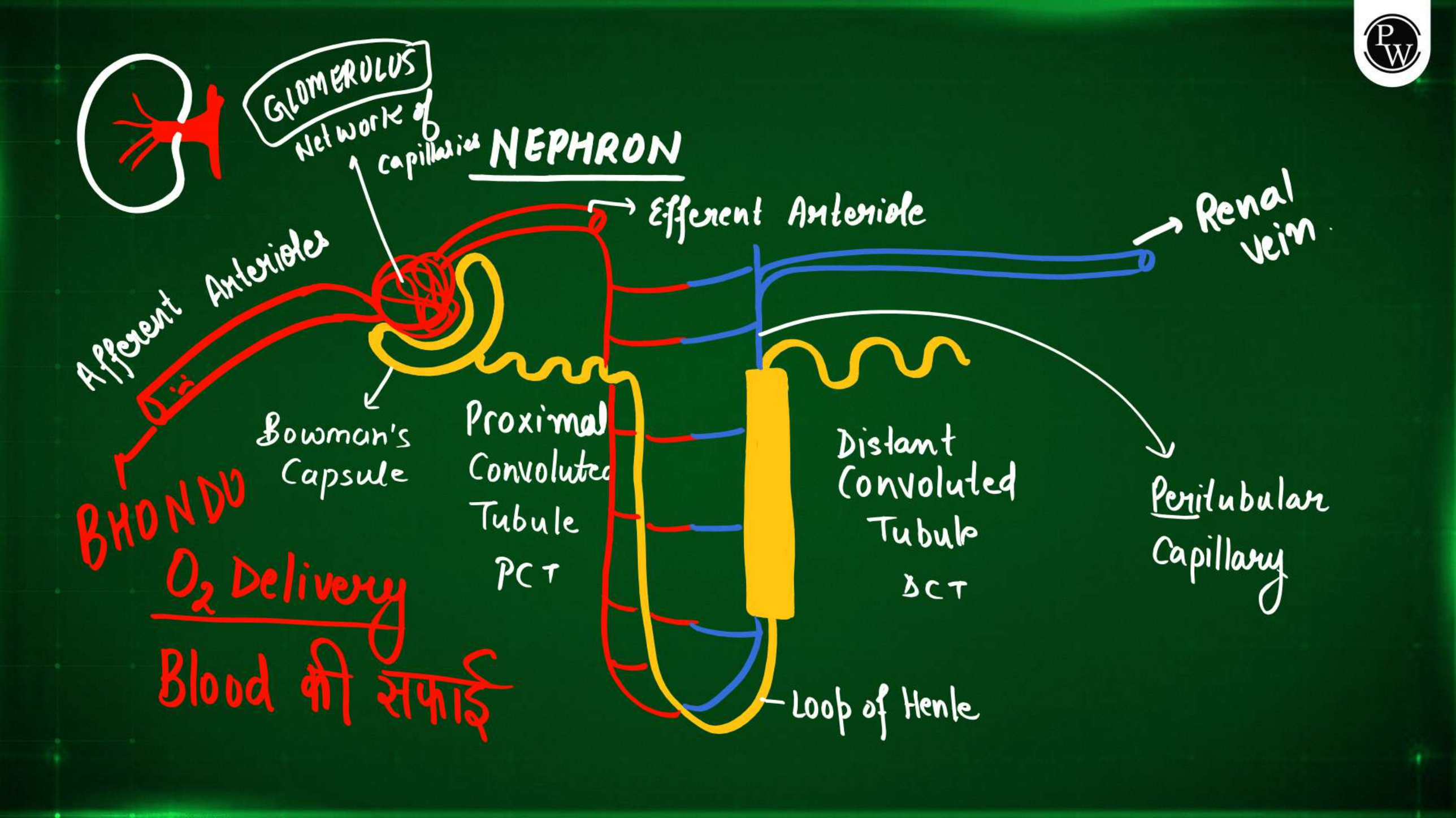
Kidney → structural & functional unit

Renal Tubules
or
NEPHRONs

10 lakh
1 Million

Kidney


Adrenal Glands
Calyx (Minor, Major)
Renal Artery
Renal Vein
Renal Cortex
Columns of Bertini
Renal Pelvis
Renal Medulla
ureter
Renal Capsule.


GLOMERULUS
Network of capillaries
NEPHRON
Efferent Arteriole
Renal vein
Afferent Arteriole
Bowman's Capsule
Proximal Convoluted Tubule
PCT
Distant Convoluted Tubule
DCT
Peritubular Capillary
Loop of Henle
BHONDU
O2 Delivery
Blood की सफाई

| Ammonia | Urea | Uric Acid |
| --- | --- | --- |
| Highly toxic | Medium toxic | Least toxic |
| Requires Maximum $H_2O$ | less amount of water | Least $H_2O$ |
| [Aquatic animals] | [Mammals] | [Birds] |
| Ammonotelic animals | Ureotelic | Unicotelic |

## QUESTIONS

**Urine is carried from the kidney to the urinary bladder by:**

**(ICSE Specimen 2023, 2013)**

A Uterus

B Urethra

C Umbilical cord

D Ureter ✓

## QUESTIONS

The pale yellow color of normal human urine: (ICSE 2021-22 Sem II)

A Chlorophyll

B Heterochrome

C Urochrome

D Haemoglobin

## QUESTIONS

**The main nitrogenous wastes formed in the human body**

**(ICSE 2021-22 Sem II)**

A. Uric acid

B. Urea

C. Ammonia

D. Cretanine

# QUESTIONS

**The animals called ammoniotelic excretes mainly:**

A Urea

B Uric acid

C Ammonia

D Both (A) and (B)

# HOMEWORK FROM VICTORY 2026 MODULE

# Thank You